Engineering Mathematics (AHT-003) - Unit 4

Unit 4 - Vector Calculus - notes in easy English, matched to the UTU syllabus.

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What this unit covers (syllabus): vectors and their differentiation; gradient, divergence, curl; directional derivative; line, surface and volume integrals; Green's, Stokes' and Gauss divergence theorems.

Key ideas: A **scalar point function** gives a number at each point (temperature); a **vector point function** gives a vector (velocity field). **Gradient** ∇φ points in the direction of fastest increase; its magnitude is the maximum rate. **Directional derivative** = ∇φ· (unit vector). **Divergence** ∇·F measures outflow per volume (source or sink); **curl** ∇xF measures rotation (curl ∇φ = 0 always). Integrals: **line** (along a curve), **surface**, **volume**. The three big theorems convert between them: **Green's** (line integral around a plane curve ↔ double integral over the region), **Stokes'** (line integral around a closed curve ↔ surface integral of curl), **Gauss divergence** (surface integral of F ↔ volume integral of div F).

Exam pointers: compute grad/div/curl of a given field (easy, mechanical marks), one directional-derivative numerical, and one theorem verification - Gauss divergence is the most asked. Remember curl grad = 0 and div curl = 0; they appear as 2-mark proofs.

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